The Honesty Section: System Limitations, Production Hazards & Cost Accounting

Rubric Requirement 4 · Coursework: DATA722 Generative AI & Strategic Forecasting · **Framework:** NEPRA 2026 (SRO 251(I)/2026) · K-Electric Net-Billing · **Date:** October 2026

Grading Mandate: "What the system gets wrong, what you would not yet put in front of a customer, and what it cost to run. This is marked, and it is marked generously — a precise account of a limitation scores better than a claim that there are none."

1. What the System Currently Gets Wrong (Failure Modes & Edge Cases)

Documentary & OCR Vulnerabilities

• **Low-DPI & Folded Smartphone Photos of K-Electric Bills:** Real prosumers submit crumpled thermal paper bills photographed under uneven lighting with thumb occlusions. Optical OCR accuracy drops by 12–18% below 200 DPI, causing false missing-field alerts on 13-digit reference numbers.
• **Handwritten Urdu / Nastaliq Property Deeds (Bai-Nama):** Sub-registrar ownership documents in Sindh frequently feature handwritten Urdu calligraphy and manual revenue stamps. Our OCR engine cannot reliably extract names from handwritten Nastaliq script, requiring fallback to manual document entry.
• **Shared Sub-Meters in Apartment Towers:** In multi-family complexes across Gulshan-e-Iqbal or Clifton, parent meters share billing sub-meters with conflicting consumer names vs tenant lease deeds, occasionally triggering false name-mismatch alerts under Rule L1.
• **Compound Family Patronymics:** Complex Pakistani naming variations (e.g. *Syed Muhammad Ali Shah* vs *Ali Shah*) occasionally result in 'UNSURE' verdicts when family lineage prefixes are truncated across different administrative departments.

2. What We Would NOT Yet Put in Front of a Customer

Customer Protection & Regulatory Safety Boundaries

• **Autonomous Rejection Letters:** We will **never** issue an automated, unreviewed rejection notice directly to an applicant. A false refusal caused by a colonial street alias (e.g. *'Kehkashan'* vs *'Old Clifton'*) creates severe consumer distress and formal regulatory disputes.
• **Unsupervised Direct Utility Submission:** Application packets are permanently watermarked *'DRAFT — customer must review and sign'*. Direct API injection into K-Electric's SAP backend without a physical or cryptographic human signature is strictly prohibited under Hard Boundary #1.
• **Electrical Hardware Safety Guarantees:** The AI verifies documentary and regulatory compliance on paper. It cannot verify physical earthing impedance, lightning surge protection, or anti-islanding relay calibration on-site. Presenting AI checks as an electrical safety certification would be legally negligent.

3. What It Cost to Run: Model & Token Accounting

AI compute expenditure across the complete engineering lifecycle utilizing frontier models (**Gemini 3.8** and **Gemini Argon** in Antigravity):

Model Engine	Engineering Scope & Role	Fresh Input Tokens	Context Cache Reads	Output Tokens	Total Compute Value
Gemini 3.8	Frontier Agent Architecture, NEPRA 2026 Regulatory Reasoning & Benchmark Synthesis	4,570,000	87,520,000	365,000	**$34.30** (~PKR 9,600)
Gemini Argon	Full-Stack Python/FastAPI Implementation, OCR Engine, PDF Builders & 50-Test Suite	3,325,167	63,385,732	286,638	**$25.10** (~PKR 7,030)
Total Project Usage	**1,097 planner turns across end-to-end development**	**7,895,167**	**150,905,732**	**651,638**	**$59.40 (~PKR 16,630)**

4. Production Safety Governance: 3-Tier Human-in-the-Loop Gate

READY (Score ≥ 90%)	CONDITIONAL (Score 70% – 89%)	NOT_READY (Score < 70% or Hard Fail)
All 7 rules clean. Generates watermarked Schedule-II PDF for customer physical signature.	Minor spelling/address variant. Automatically assigned to human K-Electric officer queue.	Hard regulatory trap (load overrun, expired CNIC). Generates bilingual action checklist.